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FINE-TUNING A CLASSIFICATION MODEL FOR FETAL ULTRASOUND PLANE IDENTIFICATION REPORT

Task : a. Fine-tuning a classification model on open-source fetal ultrasound dataset from Zenodo**.**

Models: EfficientNet-B0 and ResNet-50 (Transfer Learning + Fine-Tuning)

Github link: **https://github.com/sowbagya-049/Fetal-Ultrasound-Plane-Classifier.git**

Dataset link: **FETAL_PLANES_DB: Common maternal-fetal ultrasound images**

1. Introduction

Ultrasound imaging is widely used in prenatal care to assess fetal development through specific anatomical views such as the brain, abdomen, and femur. Identifying these views requires expertise and is time-consuming.

In this project (Task A), I fine-tuned classification models using transfer learning. Pre-trained models (EfficientNet-B0 and ResNet-50) were adapted to classify fetal ultrasound images using a publicly available Zenodo dataset.

2. Dataset

The **FETAL_PLANES_DB** dataset (Zenodo) contains over **12,000 fetal ultrasound images** collected from two hospitals and annotated by expert clinicians. The images are categorized into **6 anatomical classes** commonly used in prenatal screening.

Dataset Summary

- Total images: 12,400 (**11,921 after cleaning**)
- Patients: 1,792
- Classes: 6
- Format: Grayscale PNG
- Train/Test split: 7,129 / 5,271
- License: CC BY 4.0

Classes

- Fetal Abdomen
- Fetal Brain
- Fetal Femur
- Fetal Thorax
- Maternal Cervix
- Other

Class Imbalance

The dataset is highly imbalanced (≈7× difference).

- Largest class: **Other (36.5%)**
- Smallest class: **Fetal Abdomen (5.0%)**

This imbalance can bias the model toward dominant classes if not properly handled.

3. Data Preparation

Data Quality Checks

The dataset was validated using automated checks for corrupted files, duplicates (MD5 hashing), missing labels, inconsistent sizes, and data leakage.

Result:

- Final dataset: 11,921 images
- No corrupted or duplicate files
- No data leakage detected

Data Split

The dataset uses an official patient-level train–test split, ensuring images from the same patient do not appear in both sets. A 10% validation split was created from the training data using stratified sampling.

Split	Samples	Purpose
Train	6,186	Model training
Validation	688	Model selection
Test	5,047	Final evaluation

4. Preprocessing and Augmentation

Preprocessing

- Converted grayscale images → RGB
- Resized to 224 × 224
- Normalized using ImageNet mean and standard deviation

Training Augmentations

I applied random augmentations during training to help the model generalise better and to simulate the natural variation seen in real clinical ultrasound scans. No augmentation was applied to validation or test images.

Augmentation	Parameter	Clinical Reason
Random Horizontal Flip	50% chance	Probe can be placed from either side of the body
Random Vertical Flip	20% chance	Scan orientation varies per operator
Random Rotation	Up to 15°	Transducer angle differs between sonographers
Color Jitter	Brightness + contrast	Different machines have different gain settings
Random Affine (translate)	10% shift	Baby can move or be in different positions
Gaussian Blur	Kernel size 3	Simulates images at different scan depths
Random Crop	224 from 244px	Handles small scale differences between machines

5. Handling Class Imbalance

The dataset shows significant class imbalance, which can bias predictions.

Techniques used:

- **Weighted Random Sampling:** Oversamples rare classes
- **Class-Weighted Loss:** Penalizes errors on minority classes
- **Label Smoothing (ε = 0.1):** Reduces overconfidence on ambiguous samples

These methods improved balanced learning across all classes.

6. Model Architecture

Transfer Learning

Training deep models from scratch requires large datasets and high compute. Instead, transfer learning was used by adapting ImageNet pre-trained models to ultrasound classification. Early layers capture general features (edges, textures), which transfer well to medical images..

EfficientNet-B0

EfficientNet-B0 is a compact and efficient model that scales network depth, width, and resolution together using a fixed ratio. It achieves good accuracy with far fewer parameters than older models like VGG or ResNet. I replaced its original 1,000-class output layer with a custom head for 6 classes:

Component	Details
Backbone	EfficientNet-B0 pretrained on ImageNet
Custom Head	Dropout(0.4) → Linear(1280→512) → ReLU → Dropout(0.3) → Linear(512→6)
Total Parameters	4,666,498
Trainable (Phase 1)	1,788,342 (head + last 2 blocks)
Trainable (Phase 2)	4,666,498 (all layers)

ResNet-50

ResNet-50 is a 50-layer deep network built on residual connections. Each layer learns a small correction (residual) on top of what the previous layers already learned, which prevents the problem of vanishing gradients in deep networks. I used the same custom head as EfficientNet-B0 but attached to ResNet's 2,048-feature output.

Component	Details
Backbone	ResNet-50 pretrained on ImageNet
Custom Head	Dropout(0.4) → Linear(2048→512) → ReLU → Dropout(0.3) → Linear(512→6)
Total Parameters	24,560,198
Trainable (Phase 1)	16,016,902 (head + layer4)
Trainable (Phase 2)	24,560,198 (all layers)

Two-Phase Fine-Tuning

Training was performed in two stages:

Phase	Epochs	Description	LR
Phase 1	1–5	Train classifier head (backbone frozen)	1e-3
Phase 2	6–20	Fine-tune full model	5e-5

This approach stabilizes learning and prevents loss of pre-trained features while adapting to ultrasound data.

7. Training Setup

Setting	Value
Batch Size	32
Optimiser	AdamW (weight decay = 0.0001)
LR Scheduler	Cosine Annealing (smooth decay)
Loss Function	CrossEntropy + Label Smoothing (0.1) + Class Weights
Gradient Clipping	Max norm = 1.0
Early Stopping	Patience = 5 epochs
Platform	Google Colab T4 GPU
Framework	PyTorch 2.x

Early stopping was added so training automatically stops if the validation accuracy does not improve for 5 consecutive epochs. This saves compute time and prevents overfitting.

8. Challenges Faced

Class Imbalance: Large variation in class distribution caused bias toward dominant classes. Solution: Weighted sampling and class-weighted loss.

Noisy and Ambiguous Labels: Ultrasound images across classes can appear visually similar (e.g., brain vs thorax), and the *"Other"* class is inherently broad.
Solution: Label smoothing reduced overconfidence on ambiguous samples.

Domain Difference from ImageNet: Pre-trained models were trained on natural images, while ultrasound images are grayscale and noisy.
Solution: Two-phase fine-tuning with a low learning rate enabled smooth domain adaptation.

Practical Issues During Development: Dataset download and local hardware limitations (no GPU) caused interruptions.
Solution: Training was moved to Google Colab for better reliability and compute support.

9. Results and Visualisations

Overall Performance

Model	Test Accuracy	Best Val Accuracy	Macro AUC	Params
EfficientNet-B0	90.53%	94.91%	0.9900	4.7M

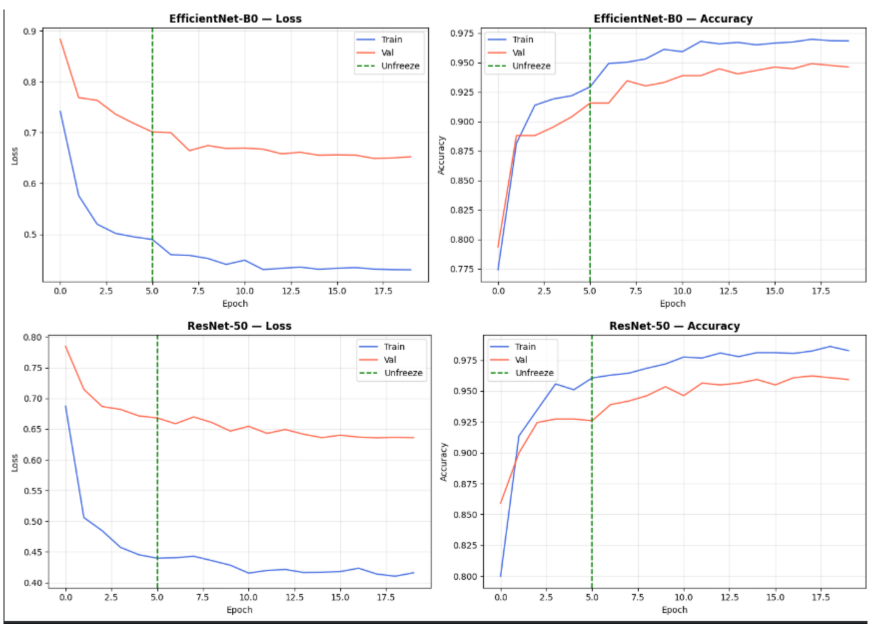
Model	Test Accuracy	Best Val Accuracy	Macro AUC	Params
ResNet-50	93.66%	96.22%	0.9913	24.6M

ResNet-50 achieved the best performance with 93.66% test accuracy and a macro AUC of 0.9913, indicating strong and reliable classification across all six fetal ultrasound classes.

Per-Class Results (ResNet-50)

Class	Precision	Recall	F1 Score	Support
Fetal Abdomen	0.8575	0.9749	0.9124	358
Fetal Brain	0.9812	0.9952	0.9882	1,472
Fetal Femur	0.7971	0.9599	0.8710	524
Fetal Thorax	0.9381	0.9409	0.9395	660
Maternal Cervix	0.9847	0.9984	0.9915	645
Other	0.9529	0.8418	0.8939	1,612

The model performed best on Fetal Brain (F1 = 0.988) and Maternal Cervix (F1 = 0.991) due to their distinctive features. The "Other" class showed the lowest recall (0.84) because of its high variability, while Fetal Femur had lower precision due to visual similarity with other structures.



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TEST RESULTS — EfficientNet-B0
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  Accuracy      : 90.53%
  Macro F1      : 0.8991
  Macro Recall  : 0.9404
  Macro Precision: 0.8784
  Macro AUC     : 0.9900
  Test Loss     : 0.6860

  Per-class report:
                 precision    recall  f1-score   support

  Fetal abdomen     0.7561    0.9771    0.8525       349
    Fetal brain     0.9814    0.9930    0.9872      1433
    Fetal femur     0.7031    0.9861    0.8209       502
   Fetal thorax     0.8663    0.9739    0.9170       652
Maternal cervix     0.9867    1.0000    0.9933       595
          Other     0.9765    0.7124    0.8238      1516

       accuracy                         0.9053      5047
      macro avg     0.8784    0.9404    0.8991      5047
   weighted avg     0.9224    0.9053    0.9039      5047


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TEST RESULTS — ResNet-50
=======================================================
  Accuracy      : 93.66%
  Macro F1      : 0.9323
  Macro Recall  : 0.9532
  Macro Precision: 0.9168
  Macro AUC     : 0.9913
  Test Loss     : 0.6580

  Per-class report:
                 precision    recall  f1-score   support

  Fetal abdomen     0.8415    0.9885    0.9091       349
    Fetal brain     0.9841    0.9923    0.9882      1433
    Fetal femur     0.8034    0.9522    0.8715       502
   Fetal thorax     0.9257    0.9555    0.9404       652
Maternal cervix     0.9916    0.9950    0.9933       595
          Other     0.9548    0.8358    0.8913      1516

       accuracy                         0.9366      5047
      macro avg     0.9168    0.9532    0.9323      5047
   weighted avg     0.9408    0.9366    0.9364      5047
```

What the Training Curves Showed

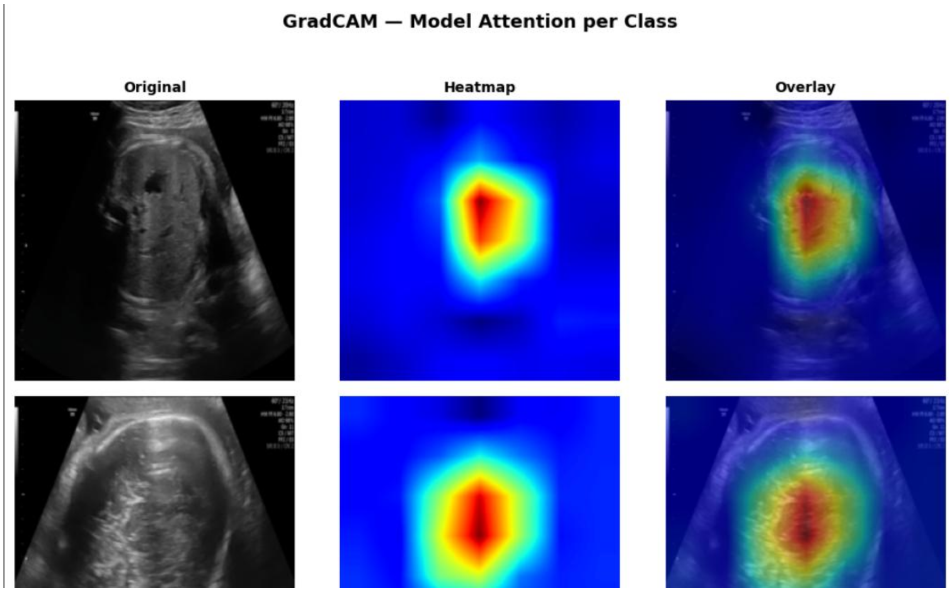
- Phase 1 (frozen backbone): Gradual accuracy improvement
- Phase 2 (fine-tuning): Significant accuracy boost (≈93% → 96%)
- Final (ResNet-50):

  Training accuracy: 98.6%

  Validation accuracy: 96.2%
- Small gap indicates good generalization (low overfitting)

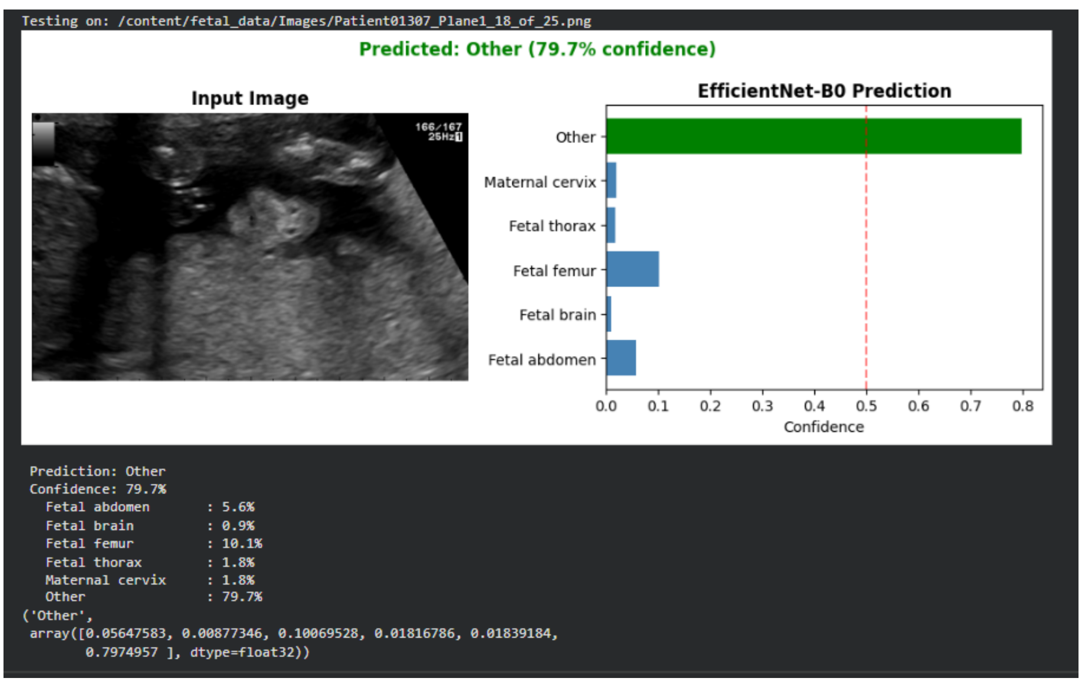
GradCAM

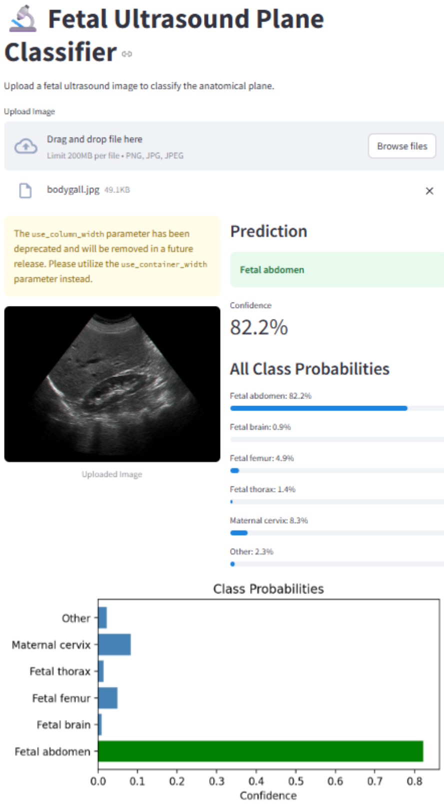
- Brain images → focused on central structures
- Femur images → focused on bone shaft
- Abdomen images → highlighted circular region
- Indicates learning of **relevant anatomical features**
- Improves **trust and explainability** for clinical use



Streamlit Result:

- Developed a Streamlit-based web app to visualize model predictions in real-time
- Allows users to upload ultrasound images and view predicted class with confidence
- Provides an interactive interface for quick testing and demonstration





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10. Key Takeaways

- Transfer learning effectively adapts ImageNet features to ultrasound images
- Two-phase fine-tuning improves stability and performance
- Handling class imbalance is critical for balanced predictions
- Label smoothing helps with ambiguous classes
- ResNet-50 outperformed EfficientNet-B0 (93.66% vs 90.53%)
- Grad-CAM confirms focus on clinically relevant regions

11. Future Directions

Short-Term Improvements

- Fine-grained brain classification (sub-plane prediction)
- Test-time augmentation for improved accuracy
- Explore Vision Transformers (e.g., Swin Transformer)

- Add uncertainty estimation (Monte Carlo Dropout)

Toward Clinical Deployment

- Real-time integration into ultrasound systems
- Federated learning for multi-hospital training with privacy
- Deploy as a lightweight app (e.g., Streamlit-based interface)

12. References

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